Extended Case 2 — Oxford Cereal

Packaging weights and inference · 307312 Applied Statistics — Masters Level

- **Start after:** sampling distributions, confidence intervals and one-mean testing in Part 3.
- **Suggested time:** 50–60 minutes.
- **Business question:** Does the reported sample provide evidence that mean box weight differs from the 368 g target?

Learning outcomes:

- Distinguish process standard deviation from the standard error of a mean.
- Calculate a Z test and interval from reported summary statistics.
- Explain what a failure to reject can and cannot establish.

1. The packaging manager's brief

- Oxford fills cereal boxes labelled **368 grams**.
- Underfilling concerns customers; overfilling uses extra product.
- **Source:** *Basic Business Statistics*, ninth edition, Chapters 7 (p. 232) and 9 (pp. 296–303; numerical example p. 302).
- **Reported evidence:** 25 boxes, sample mean 372.5 g, assumed known population SD 15 g.
- **Unit:** grams per box; the supplied CSV has one summary row, not 25 box records.
- **Limit:** raw weights, collection dates and individual underfill counts are unavailable.
- The publisher's separate `CEREALS` cost/nutrition dataset is not Oxford packaging data.

2. Import the reported inputs

```
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
oxford <- read.csv(file.path(data_dir, "business", "extended-oxford-summary.csv"))
knitr::kable(oxford)
```

n	sample_mean_g	population_sd_g	target_mean_g	alpha
25	372.5	15	368	0.05

```
n <- oxford$n[1]
xbar <- oxford$sample_mean_g[1]
sigma <- oxford$population_sd_g[1]
mu0 <- oxford$target_mean_g[1]
alpha <- oxford$alpha[1]
stopifnot(n == 25, sigma == 15, mu0 == 368, xbar == 372.5)
```

- `data/business/extended-oxford-summary.csv`; guide: `data/business/README.md`.
- `sigma` is assumed known in the textbook example; it is not estimated from these 25 boxes.
- A Z procedure requires the stated known SD and an appropriate sampling/normal model.

3. Separate variation between boxes and between sample means

```
se <- sigma / sqrt(n)
knitr::kable(data.frame(sample_size = n, sample_mean_g = xbar,
  difference_from_target_g = xbar - mu0, process_sd_g = sigma,
  standard_error_g = se), digits = 2)
```

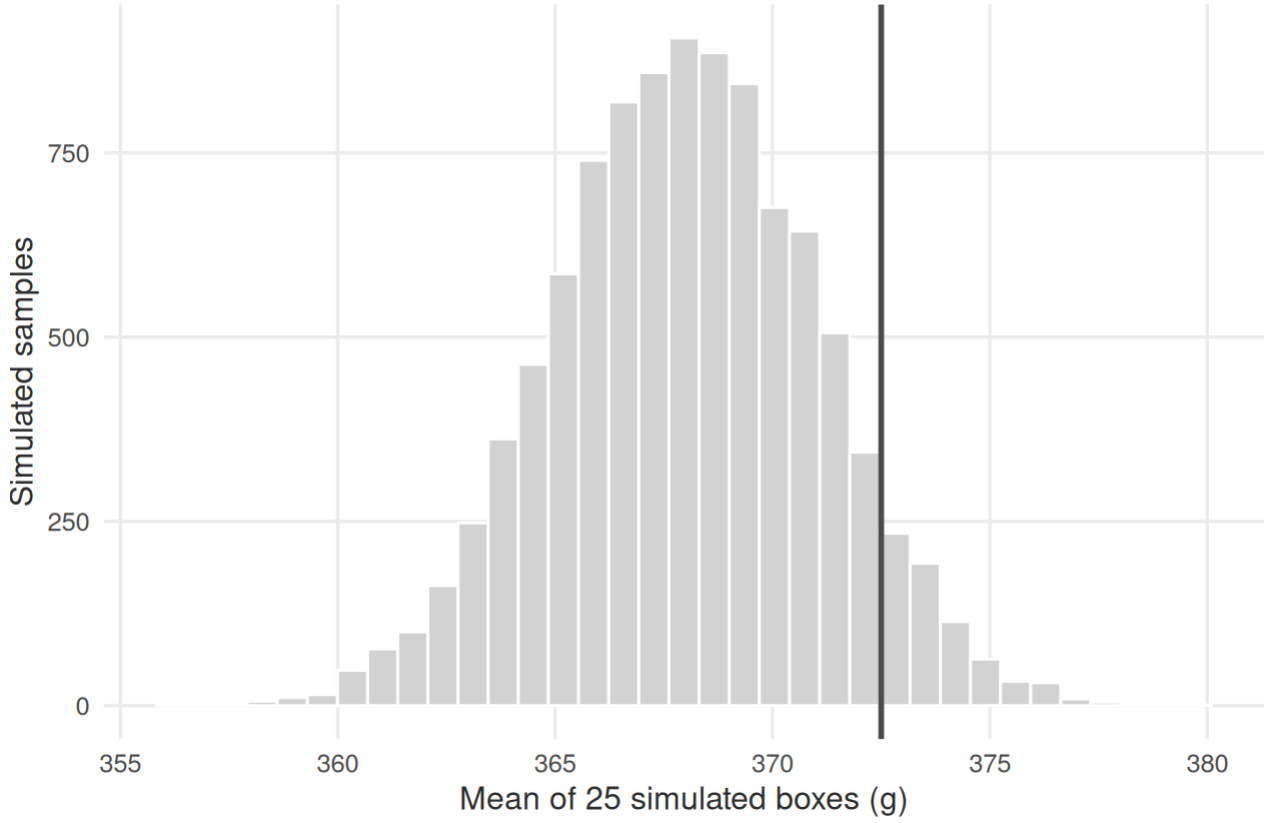
sample_size	sample_mean_g	difference_from_target_g	process_sd_g	standard_error_g
25	372.5	4.5	15	3

- `sigma = 15` g describes variation between individual boxes under the model.
- `se = 3` g describes variation between means of independent samples of 25 boxes.
- Increasing sample size reduces the standard error; it does not mechanically reduce process variation.

4. See the sampling distribution

- This simulation generates **hypothetical** normal-process weights under a 368 g mean and 15 g SD.
- Its values are not observed Oxford measurements.

```
set.seed(42001)
sim_means <- replicate(10000, mean(rnorm(n, mean = mu0, sd = sigma)))
ggplot(data.frame(sample_mean = sim_means), aes(sample_mean)) +
  geom_histogram(bins = 35, fill = petra_accent, colour = "white") +
  geom_vline(xintercept = xbar, colour = petra_ink, linewidth = 1) +
  labs(x = "Mean of 25 simulated boxes (g)", y = "Simulated samples")
```



- `set.seed()` makes the simulation reproducible.

- `replicate()` repeats the hypothetical sampling experiment.
- The vertical line marks the reported sample mean, 372.5 g.

5. Test the process mean

- **Null:** the population mean is 368 g.
- **Alternative:** the population mean differs from 368 g.
- **Decision threshold:** 5%, two-sided.

```
z <- (xbar - mu0) / se
p_value <- 2 * pnorm(-abs(z))
ci <- xbar + c(-1, 1) * qnorm(1 - alpha / 2) * se
knitr::kable(data.frame(z = z, p_value = p_value,
  lower_mean_g = ci[1], upper_mean_g = ci[2], reject = p_value < alpha), digits = 4)
```

z	p_value	lower_mean_g	upper_mean_g	reject
1.5	0.1336	366.6201	378.3799	FALSE

- `pnorm()` gives the normal tail probability; the two tails account for deviations in either direction.
- The result is **Z = 1.50**, with p about **0.134**.
- The 95% interval for the process mean is approximately **366.62–378.38 g**.
- The evidence is insufficient to reject the 368 g mean at 5% under the stated assumptions.
- This does not prove exact centering, individual-box compliance or a low underfill rate.

6. Independent packaging review

1. Explain why `sd()` or `t.test()` on the single summary row cannot analyse 25 individual boxes.
2. Recalculate the test and interval for a **hypothetical** sample mean of 375 g, holding the other inputs fixed.
3. Compare the standard errors for hypothetical independent sample sizes 25 and 100 with the same known process SD.
4. Explain why a narrow interval for the mean does not guarantee that individual boxes meet a weight requirement.
5. Describe Type I and Type II errors as unnecessary process intervention and missed process changes.
6. Specify the raw measurements and sampling records needed to investigate underfilling and process stability.

```
# Independent task: change the summary inputs and label the scenario hypothetical.
# Use (sample_mean - target) / (known_sd / sqrt(sample_size)).
```

Submission

- One comparison table for the reported and hypothetical scenarios.
- A sampling-distribution chart with clearly labelled simulated values.
- Five briefing bullets covering evidence, decision, assumptions, limits and further data.